

POST-GRADUATE RESEARCH FORUM



Dynamic Knowledge Graph Discovery from Academic PDFs

Hypergraphs · 62 Discovery Operators · Bridge Analogies · Graph-RAG · Causal V2

System	Core	LLM Backends	Status
DynamicKnowledgeGraphs	C++17 · FastAPI · Kuzu	OpenAI / Gemini	Active · V2 In Progress



01 — MOTIVATION

Academic Knowledge Remains Trapped in Unstructured Text

THE GAP

Standard RAG pipelines retrieve text chunks — they do not reason over *structure*.

HIGHER-ORDER RELATIONS

Important findings involve multiple entities simultaneously — not just subject–predicate–object triples.

CROSS-DOMAIN INSIGHT

Valuable analogies and bridging hypotheses sit *between* papers, not inside any single one.

PDF

— standard pipeline stops here —

Hypergraph
→ Discovery
→ Reasoning

02 — SYSTEM OVERVIEW

Three Active Code Paths, One Unified Architecture

Primary

kg CLI

C++17 pipeline — extraction, QC, indexing, discovery, rendering, reporting.

```
./build/bin/kg run -i  
docs/ -p all
```

Interactive

Graph-RAG Tool

FastAPI + Kuzu backend — natural-language querying over the extracted graph.

```
python graph_rag_server.py
```

Lightweight

Python Port

Azure OpenAI workflow — extraction only, no QC, discovery, or reports.

```
python kg_extractor.py  
file.pdf
```

Design choice: performance-sensitive graph logic in C++ · user-facing interaction served through the web stack

03 / 15

03 — PIPELINE STAGES

Six Stages from PDF to Interactive Insight

- 1

Knowledge Extraction
PDF → chunks → LLM → entities + hyperedges
- 1.5

Preprocess (opt)
Relation normalisation · alias merging
- 1.75

Quality Control (default on)
5-level cleaning pass
- 1.9

Ontology Classification (opt)
class / instance labelling
- 2

Index Building
HypergraphIndex · s-components (s=2,3,4)
- 3

Knowledge Discovery
62 operators · typed Insight objects

- 4

Visualisation
graph.html · graph_augmented.html · graph_rag.html · graph.dot
- 5

Report Generation
report.md · report.html · pattern_library.json
- 6

Manifest
manifest.json · artifact index · timestamps

RESUME SUPPORT
`kg run -f 3 -d runs/run_20260424_120000`
Resume from any stage · preserves all prior artifacts

Each stage saves to `runs/run_YYYYMMDD_HHMMSS/`

04 — DATA MODEL

Hyperedges Preserve What Binary Triples Lose

Standard Graph — Triples

```
// Forces binary decomposition:
(method) → uses → (dataset)
(method) → scored_by → (metric)
(metric) → has_value → (score)
// Relation is lost across 3 edges
```

Hypergraph — N-ary Hyperedge

```
// Single hyperedge, four participants:
[method, dataset]
  -evaluated_with→
[metric, score]
```

WHAT HYPEREDGES ENABLE

- Compositional n-ary relations preserved
- Richer path reasoning through the index
- Motif discovery on multi-participant patterns
- s-component analysis ($s = 2, 3, 4$)

S-COMPONENT INDEXING

Nodes sharing $\geq s$ hyperedges form a component. $s=2,3,4$ computed at stage 2. Used by discovery operators to reason about dense structural clusters.

05 — QUALITY CONTROL

Five-Level Quality Control Before Discovery Begins

LEVEL	STAGE	WHAT IT DOES	COST
1	Rule-based filtering	Short labels, stopwords, pure numbers, extraction artifacts	~1ms / 1K
1.2	Label simplification	Strips filler phrases: "the process of", "a method for"	~1ms / 1K
1.5	Semantic deduplication (opt)	"ML" ↔ "machine learning" — LLM merges near-duplicate nodes	LLM call
2	Statistical filtering	Degree-based outlier removal · graph connectivity analysis	~50ms / 1K
3	LLM validation (opt)	Batch VALID/INVALID classification of suspicious entities	~3s / 50

06 — DISCOVERY ENGINE

62 Discovery Operators Across Three Families

COMBINATORIAL

Recurring structure · hierarchy · dense subgraphs

- Motif
- K-Core
- Community
- Core-Periphery
- Meta-Path
- K-Truss
- Centrality
- Transitive Closure
- Meta-Pattern
- Taxonomy

EXPLORATORY

Missing links · bridges · cross-domain traversal

- Bridge
- Path Rank
- Causal Chain
- Explanatory Chain
- Citation Impact
- Completion
- Domain Bridge
- Diffusion
- Translation Gap
- Long Chain

TRANSFORMATIONAL

Reframing · contradiction · anomaly · uncertainty

- Contradiction
- Hypothesis 1/2/3
- Schema Repair
- Feedback Loop
- Benchmark Dep.
- Counterfactual
- Anomaly
- Confounder
- Evidence Debt

The KG is not the final product — it is the substrate for automated discovery.

07 — NOVEL CONTRIBUTIONS

Four Novel Contributions at the Core of the Framework

Bridge-Mediated Analogical Reasoning

Graph bridges as mediators for cross-domain analogies. Structure-based — not text similarity or embedding arithmetic.

84% precision

Pattern Library System

Reusable structural templates exported as structured JSON. Enables cross-study comparison and template-based discovery.

73 patterns extracted

Bias Audit Metrics

Gini-coefficient representation inequality analysis during knowledge construction, not just downstream tasks.

Gini 0.38 on corpus

Community-Aware Recommendations

Novelty-weighted scoring balances within-community familiarity and cross-community discovery.

88% relevance (manual)

08 — BRIDGE-MEDIATED ANALOGY

Bridge Entities Enable Cross-Domain Analogical Reasoning

```
// Algorithm 1: Bridge-Mediated Analogy for each
bridge b in B:
  communities ← get_communities(b)
  for each Ci in communities:
    patterns[Ci] ← extract_patterns(b, Ci)
  for each pair (Ci, Cj):
    shared ← patterns[Ci] ∩ patterns[Cj]
    if confidence(shared) > θ:
      A ← A ∪ construct_analogy(b, shared)
```

EXAMPLE — BRIDGE: "KNOWLEDGE GRAPH"

ML community: `neural_network` → uses → optimization
Symbolic AI: `knowledge_graph` → uses → reasoning
Hypothesis: Gradient-based optimisation may apply to KG reasoning.

CONFIDENCE SCORING

score = (complexity + pattern_strength) / 2
complexity = min(1, |entities| / 8)
pattern_strength = min(1, |shared| / 3)

100+ cross-domain analogies

84% precision (50 validated)

09 — EVALUATION

Evaluation: 22 Papers, 297 Entities, 113 Insights

84%

bridge analogy precision

70%

cross-domain pattern transferability

0.38

Gini coefficient (moderate fairness)

20x

dev efficiency vs manual approach

TEST CORPUS

22 PDF papers

297 entities

182 relations

113 insights

PATTERN LIBRARY RESULTS

45 motifs

27 k-truss

73 total patterns

This is not "big data" scale. It is a research-scale corpus designed to test whether discovery methods produce useful, interpretable outputs.

10 — GRAPH-RAG INTERFACE

Natural Language Querying via Graph-RAG

BACKEND STACK

- FastAPI server · REST + WebSocket
- Kuzu graph database (in-process)
- sentence-transformers 384-d (local)
- Azure OpenAI 1536-d (cloud)

THREE VIEWER OUTPUTS

graph.html — 3D force-directed · Louvain clustering
graph_augmented.html — insight nodes overlaid on
base
graph_rag.html — NL Q&A chat over the graph

```
# Load any completed run's graph.json
cd src/graph_rag_tool/backend
python graph_rag_server.py

# Open
http://localhost:8000 # 3D viewer + chat
http://localhost:8000/docs # API reference
```

```
# Example NL query:
"What connects machine learning
to symbolic reasoning in this corpus?"
```

MULTI-HOP REASONING

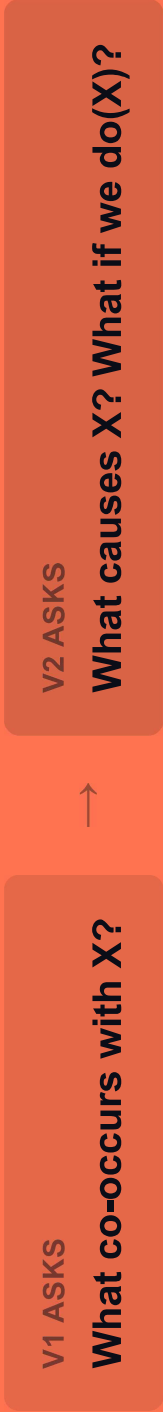
The augmented viewer overlays discovered insight nodes onto the graph — showing not just what entities exist, but what structural patterns were found between them.

V2 DESIGN DIRECTION

V2: From Association to Causal Mechanism

V1 represents *correlations*.

V2 aims to represent *causation, confounding, and intervention*.



11 — V2 ARCHITECTURE

Pearl's Causal Hierarchy Embedded in the Pipeline

LEVEL 1 — ASSOCIATION (V1)

$$P(Y \mid X)$$

What fraction of papers mention X and Y together?

LEVEL 2 — INTERVENTION (V2)

$$P(Y \mid \text{do}(X))$$

What happens if we force X to a value? Backdoor / frontdoor adjustment.

LEVEL 3 — COUNTERFACTUAL (V2)

$$P(Y_x \mid X=x', Y=y')$$

What would have happened if X had been different?

CAUSAL METADATA PER EDGE

```
"causal_type": "direct_cause",  
"direction": ["C02", "acidification"],  
"evidence_type": "experimental",  
"mechanism": "carbonic acid formation",  
"confounders": ["natural_variability"],  
"confidence": 0.87
```

NEW DISCOVERY OPERATORS (V2)

causal_chain_detection confounder_identification
mediator_analysis collider_detection
intervention_simulation

Phase 2 implemented · Phases 3–8 in progress

12 — OPEN PROBLEMS

Known Limitations and Open Problems

LLM EXTRACTION NOISE

Over-generates low-value relations. Generic entities ("Fig", "Table", single-letter labels) pollute the graph. QC helps but is not a complete solution — garbage in still means noise out.

VALIDATION COST

Manual validation of bridge analogies and causal claims is expensive. No automated ground-truth evaluation dataset for analogy discovery currently exists.

SCALABILITY CEILING

Current implementation targets graphs under 100K nodes. Scaling to larger corpora requires additional engineering — particularly for the discovery and indexing stages.

CAUSAL EXTRACTION RELIABILITY

LLMs may hallucinate causal claims or miss subtle confounders. The V2 design proposes multi-model ensembles, confidence calibration, and human-in-the-loop review for low-confidence claims.

"The project is promising — but not solved."

14 / 15

13 — CONCLUSION

A Knowledge Discovery System Built on Extracted Graphs

Distinctive features:

- Hypergraph representation of n-ary academic relations
- 62 operator-driven discovery across three families
- Bridge-mediated analogical reasoning — structural, not text-based
- Pattern libraries + fairness-aware bias auditing
- Interactive Graph-RAG with 3D visualisation
- V2: Pearl's causal hierarchy from association to intervention

FUTURE WORK

- Automated evaluation dataset for analogy discovery
- Pattern similarity search and cross-graph matching
- Scalability to >1M node corpora
- Causal DAG construction + do-calculus (V2 phases 4–5)
- Human-in-the-loop analogy refinement

github.com/andrediasbarbosa/DynamicKnowledgeGraphs
C++17 · FastAPI · Kuzu · OpenAI / Gemini · Apache 2.0